Ivan **Di Liberti**

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Logic, Foundations of Mathematics, Category theory

Doctrines, (2-)toposes, sketches, accessible and locally presentable (2-)categories, semantics of type theory, formal category theory.

Appointments

Assistant Professor (biträdande lektor) — *Gothenburg, Sweden*

University of Gothenburg, Department of Philosophy, Linguistics and Theory of Science — *since Jan 2025*

Coordinator of the Master in Logic at Gothenburg University

Postdoc — *Gothenburg, Sweden*

University of Gothenburg, Department of Philosophy, Linguistics and Theory of Science — *Oct 2023 - Dec 2024*

Project: Logical Foundations of Circularity
Principal Investigator: Graham Leigh

Postdoc — *Stockholm, Sweden*

Stockholm University, Department of Mathematics — *Jan 2022 - Sept 2023*

Project: Cubical methods for computer formalization of mathematics
Principal Investigator: Anders Mortberg

Postdoc — *Prague, Czech Republic*

Czech Academy of Sciences, Institute of Mathematics — *Jul 2020 - Dec 2021*

Project: Abstract Convergence Schemes And Their Complexities
Principal Investigator: Wiesław Kubiś

Education

Ph.D. in Mathematics — *Brno, Czech Republic*

Masaryk University — *Feb. 2017 - Sept. 2020*

Dissertation: The Scott Adjunction
Advisor: Jiří Rosický

MSc in Mathematics — *Pisa, Italy*

Università di Pisa — *Mar. 2015 - Dec. 2016*

Dissertation: A survey on Barr's Theorem
Advisors: Nicola Gambino, Alessandro Berarducci

BSc in Mathematics — *Pisa, Italy*

Università di Pisa — *Sept. 2011 - Mar. 2015*

Dissertation: Definable groups in o-minimal structures
Advisor: Alessandro Berarducci

Awards and Grants

2026 **Host (co)applicant for CSC scholarship of Xiao Hu (approx. 20k euro)**,
Research project selected for support by the China Scholarship Council (CSC), which awarded a fully funded doctoral scholarship to Xiao Hu to conduct research under my supervision. The CSC is China's principal governmental agency for international academic funding and supports highly competitive overseas doctoral training and research collaborations.

2025 **Mobility Grant (co)applicant in the supervision team of Robin Jurde (10k euro.)**,
Our project LANGSPEC: Locally Presentable Categories for Programming Language Specification was selected for support by CNRS Mathematics (Insmi) as an International Research Project (IRP), a competitive programme funding long-term international scientific collaborations. The project supports a three-year research partnership between France and Sweden and received dedicated CNRS funding for researcher mobility and scientific events.

2021 **Masaryk University Vice-rector's award for excellent results in doctoral studies**,
Forty graduates of doctoral programmes and their lecturers have recently accepted from Vice-rector for research and doctoral studies Šárka Popíšilová an award for excellent achievements in their doctoral studies at Masaryk University.

Publications

2024 **Bi-accessible and bipresentable 2-categories**, *j/w Axel Osmond*
Applied Categorical Structures. arXiv:2203.07046.

2024 **Adjoint functor theorems for lax-idempotent pseudomonads**, *j/w Arkor, Loregian*
Theory and Application of Categories. Vol. 41, 2024, No. 20, pp 667-685.

2024 **Context, Judgement, Deduction**, *j/w Greta Coraglia*
Accepted by Proceedings of CatMi 2023. arXiv:2111.09438.

2024 **Coalgebras for Vietoris and monadicity**, *j/w Marco Abbadini*
The Journal of Symbolic Logic. 10.1017/jsl.2024.14.

2024 **The geometry of coherent topoi and Ultrastructures**,
Conditionally accepted by The Journal of Symbolic Logic. arXiv:2211.03104.

2023 **Formal Model Theory & Higher Topology**,
Mathematical Logic Quarterly. 10.1002/malq.202300006.

2023 **KZ-pseudomonads and Kan Injectivity**, *j/w Lobbia, Sousa*
Theory and Applications of Categories, arXiv:2211.00380.

2022 **Accessibility and Presentability in 2-categories**, *j/w Fosco Loregian*
Journal of Pure and Applied Algebra, 10.1016/j.jpaa.2022.107155.

2022 **Exponentiable Grothendieck categories in flat Algebraic Geometry**, *j/w Julia Ramos González*
Journal of Algebra, 10.1016/j.jalgebra.2022.03.040.

2022 **Enriched locally generated categories**, *j/w Jiří Rosický*
Theory and Applications of Categories, Vol. 38, 2022, No. 17, pp 661-683.

2021 **General facts on the Scott Adjunction**,
Applied Categorical Structures, 10.1007/s10485-021-09666-6.

2021 **Towards Higher Topology**,
Journal of Pure and Applied Algebra, 10.1016/j.jpaa.2021.106838.

2021 **Functorial Semantics for Partial Theories**, *j/w Loregian, Nester, Sobocinski*
Symposium on Principles of Programming Languages (POPL21), 10.1145/3434338.

2020 **Codensity: Isbell duality, pro-objects, compactness and accessibility**,
Journal of Pure and Applied Algebra, 10.1016/j.jpaa.2020.106379

2020 **Gabriel-Ulmer duality for topoi and its relation with site presentations**, *j/w Julia Ramos González*
Applied Categorical Structures, 10.1007/s10485-020-09605-x

2019 **Weak saturation and weak amalgamation property**,
Journal of Symbolic Logic, 10.1017/jsl.2019.45

2018 **Homotopical Algebra is not concrete**, *j/w Fosco Loregian*
Journal of Homotopy and Related Structures, 10.1007/s40062-018-0197-3

Preprints

2026 **Beth companions of finitary essentially algebraic theories**, *j/w Luca Reggio*
arXiv:2609.03601. Submitted to Journal of the London Mathematical Society in October 2026.

2026 **Conceptual completeness for subgeometric logics**, *j/w Tarantino and Ye*
arXiv:2607.02250. Submitted to American Journal of Mathematics in July 2026.

2026 **Polarities, voltages, and capacitors: a categorical approach to hulls, envelopes, and completions**, *j/w Lupini*
arXiv:2603.13569. Submitted to Journal of Pure and Applied Algebra in May 2026.

2026 **Craig interpolation for for subgeometric logics**, *j/w Ye*
arXiv:2601.11221. Submitted to Journal of Symbolic Logic in Jan 2026.

2025 **Classifying infinity Topoi via weighted limits**, *j/w Meadows*
arXiv:2512.15613. Submitted to Journal of the London Mathematical Society in Jan 2026.

2025 **Logics and Concepts in the 2-category of Topoi**, *j/w Ye*
arXiv:2504.16690. Submitted to Annals of Pure and Applied Logic in Spring 2025.

2024 **Topoi with enough points**, *j/w Rogers*
arXiv:2403.15338. Working on resubmission.

2024 **Sketches and Classifying Logoi**, *j/w Lobbia*
arXiv:2403.09264. Submitted to Advances in Mathematics in Spring 2024.

Teaching

Topos Theory, *2026*
Lecturer, Master in Logic, Gothenburg University, Gothenburg, Sweden.

Introduction to Categorical Logic, *2025*
Lecturer, Tsinghua Logic Summer School, Tsinghua University, Beijing, China. j/w Lingyuan Ye

Semantics of (dependent) type theory, *2025*
Lecturer, Master in Logic, Gothenburg University, Gothenburg, Sweden.

Set Theory, *2025, 2026*
Lecturer, Master in Logic, Gothenburg University, Gothenburg, Sweden.

Model Theory, *2024, 2025, 2026*
Lecturer, Master in Logic, Gothenburg University, Gothenburg, Sweden.

Category Theory, *2024, 2025, 2026*
Lecturer, Master in Logic, Gothenburg University, Gothenburg, Sweden.

Functorial Semantics, *2024, 2025*
Lecturer, Master in Logic, Gothenburg University, Gothenburg, Sweden.

Introduction to Topos Theory, *2024*
Lecturer, PhD course, Gothenburg University, Gothenburg, Sweden.

Categorical Logic, *2023*
Lecturer, PhD course, Stockholm University, Stockholm, Sweden.

Sheaves, Manifolds and Cohomology, *2021*
Lecturer, Master in Mathematics, Charles University, Prague, Czechia. j/w Andrea Gagna.

Category Theory, *2020*
Lecturer, Excellence course for researchers, Institute of Mathematics, Czech Academy of Sciences, Prague, Czechia.

Topology, *2019, 2020*
Teaching Assistant, Bachelor in Mathematics, Masaryk University, Brno, Czechia.

Topics in Category Theory, *2018, 2019*
Lecturer, Master in Mathematics, Masaryk University, Brno, Czechia. j/w John Bourke.

Rings and Modules, *2017*
Teaching Assistant, Bachelor in Mathematics, Masaryk University, Brno, Czechia.

Supervision

PhD students,

- Robin Jourde: cosupervision of his PhD studies on Skeletal semantics. Savoie Mont Blanc University. *Ongoing*.
- Xiao Hu: supervision for a project on topos-theoretic approaches to dividing lines, Sichuan University. *Ongoing*.
- Errol Yuksel: supervision for a project on topos-theoretic Galois theory, Stockholm University. *2023-2026*.

Master students,

- Donglai Xie, Categorical Models of Dependent Type Theory with Morphisms between Types. Master in Logic. Gothenburg University.
- Aditya Prakash, Terms are Conditions: Syntactic Portraits of Properties of Algebras. Master in Logic. Gothenburg University. .
- Yiannis Hadjicharalambous, Semirings for Database Provenance. Master in Logic. Gothenburg University. Cosupervised with Phokion Kolaitis. .
- Jonathan Osser, A 2-Sketchy Approach to Type Theory. Master in Logic. University of Amsterdam.
- Alessio Zaninotto, Boolean-valued models of Set Theory. Master in Logic. Gothenburg University.
- Guido Boccali, Classifying topoi, generic semantics, and Morita theory. MSc (Philosophy). University of Torino.

Bachelor students,

- Zike Liu: Forcing with Topoi, Philosophy, Renmin University. *Ongoing*.
- Vít Jelínek: Categorical View of Monads in Computer Science, Computer Science, Masaryk University.
- Kento Okura: Sheaf Semantics over the Étale Site, Mathematics, Vienna University.
- Jonathan Osser: Exploring the Mathematical Multiverse, a brief overview of Elementary Topoi and their Internal Language, Mathematics, Stockholm University. Cosupervised with Anders Mortberg.

Other forms of supervision,

- Adam Donadille: ENS internship for six months on topos theoretic approaches to forcing. MSc (CS). ENS de Lyon.
- Lingyuan Ye: project for his Master studies on Givant categoricity theorem. MSc (M). Institute of Logic, Language and Computation, Amsterdam.

SIGPLAN long-term mentorship program,

The SIGPLAN long-term mentorship program. helps junior researchers and master students to form long-term connections in the programming languages community, and to access the perspectives of researchers from other institutions. From Summer 2021 to Summer 2025 I mentored in this project.

Organising roles and Scientific Committees

Programme and Scientific Committees,

- GothCat (2026-now)
- **Types 2026**
- ItaCa 2025 (chair)
- Bohemian Logical & Philosophical Café (2021)
- ItaCa Fest (2020-now)
- ItaCa 2019
- Peripatetic Seminar on Sheaves and Logic 103 (2018)

Organising Roles,

- Gothenburg Cyclothon 2025
- **Logic Colloquium 2024**
- Bohemian Logical & Philosophical Café (2021)
- Masaryk University Algebra Webinar (2020)
- Peripatetic Seminar on Sheaves and Logic 103 (2018)

2026 **PhD thesis Committee member for Thiago Alexandre**,

2026 **Discussion leader for the half-way seminar of my PhD student of Lorenzo Perticone**,
At Chalmers University of Technology, the discussion leader for a PhD half-way seminar serves as an independent senior reviewer of the student's work. They critically assess the manuscript and lead a structured discussion, probing methodology, contributions, and research direction. Their role is to provide rigorous, constructive feedback to support the successful completion of the PhD.

2025-2029 **Supervisory committee member for the Ph.D of Laura Stéphy Constant**,
Responsibilities include evaluating annual progress reports and thesis presentations, participating in committee meetings (initially online, later in person), and contributing to the final thesis evaluation and defense at Louvain-la-Neuve. The role also involves potential research visits and academic exchanges between institutions during the doctoral period.

2026 **Reviewer for Grant evaluation for the Czech Science Foundation (GAČR)**,
I served as a *foreign* reviewer for the Czech Science Foundation (GAČR) in evaluating an EXPRO five-year GAČR grant.

2026 **Reviewer for Grant evaluation for the Czech Science Foundation (GAČR)**,
I served as a *foreign* reviewer for the Czech Science Foundation (GAČR) in evaluating a Standard three-year GAČR grant.

2025 **Reviewer for Grant evaluation for the Czech Science Foundation (GAČR)**,
I served as a *foreign* reviewer for the Czech Science Foundation (GAČR) in evaluating a Standard three-year GAČR grant.

Community

ItaCa, Promoting Committee,
ItaCa is the Italian community of category theory. The project was founded by Giuseppe Metere in 2019.

- (2019) I was invited to join the **Scientific Committee** of ItaCa 2019 (University of Milano), the first official meeting of the community. Since this invitation I have been an active member of ItaCa.
- (2019 - now) I joined the **Program Committee** of ItaCa Fest. ItaCa Fest is a webinar aimed to gather the community of ItaCa.
- (2021 - 2026) I am one of the members of the Promoting Committee, this is the coordinating committee of the entire project.
- (2025) Chair of the **Scientific Committee** of ItaCa 2025 (University of Milano).
- (2025 - now) Co-chair of the **Mentorship program** of ItaCa.

Referee Activity,

- Advances in Mathematics
- The Journal of Symbolic Logic
- Annals of Pure and Applied Logic
- Journal of Pure and Applied Algebra
- Bulletin de la Société Mathématique de France
- Canadian Journal of Mathematics
- Theory and Applications of Categories
- Applied Categorical Structures
- Mathematical Structures in Computer Science
- Archive for Mathematical Logic.
- Algebra Universalis
- Compositionality
- Symposium on Logic in Computer Science
- Conference on Mathematical Foundations of Programming Semantics
- International Conference on Formal Structures for Computation and Deduction
- Annual Conference on Computer Science Logic.

MathOverflow, top 0.1 % user,
MathOverflow is a mathematics question-and-answer (Q&A) website, which serves as an online community of researchers in mathematics. It allows users to ask questions, submit answers, and rate both, all while getting merit points for their activities. It is a part of the Stack Exchange Network and has an internal rating system based on the reputation gained by the users within the website. Here is my profile webpage.

The Scandinavian Logic Society, Member,
Since late 2024 I am a member of the Scandinavian Logic Society.

RIFO — Swedish Association of Members of Parliament and Researchers, Member,
Since fall 2026 I am a member of RIFO. RIFO is a forum promoting dialogue and knowledge exchange between researchers and the Swedish Parliament (Riksdag). Connects scientific and technological developments with parliamentary work through seminars, discussions and study visits.

Invited Visits

2026 **Institute de recerche et informatique fondamentale (IRIF), Paris**,
Collaboration with prof. Paul-André Mellies, Sam van Gool, Umberto Tarantino.

2025 **University of Stockholm, Stockholm**,
Invited speaker at local research seminar by Peter Lumsdaine.

2025 **Tsinghua University, Beijing**,
Guest Lecturer for the Tsinghua Logic Summer School.

2025 **University of Milan, Milan**,
Invited speaker at local research seminar by Luca Reggio.

2025 **University of Bologna, Bologna**,
Invited speaker at local research seminar by Martino Lupini.

2024 **Tallinn University of Technology, Tallin**,
Collaboration with Loregian and Arkor.

2022 **Manchester University, Manchester**,
Collaboration with Nicola Gambino.

2022 **Université catholique de Louvain, Louvain**,
Collaboration with Julia Ramos Gonzalez.

2022 **Université Paris Nord, Paris**,
Collaboration with Morgan Rogers.

2021 **Institute de recerche et informatique fondamentale (IRIF), Paris**,
Collaboration with prof. Paul-André Mellies and Axel Osmond.

2019 **Institute de recerche et informatique fondamentale (IRIF), Paris**,
Collaboration with prof. Sam van Gool.

2019 **Antwerp University**,
Collaboration with Julia Ramos Gonzalez.

2018 **Max Plank Institute, Bonn**,
Collaboration with Fosco Loregian.

Selected Invited Talks

2026 **Lindstrom theorem for fragments of geometric logic**,
Ultracategories and Stone Duality Workshop, Paris, France

2026 **Fragments of geometric logic**,
Seminar lecture, local seminar, University of Ljubljana, Slovenia

2025 **From Tarski to Lawvere, algebraic aspects of predicate logic**,
16th Tsinghua Logic Colloquium, Beijing, China.

2024 **Logics in the 2-category of topoi**,
Categorical Logic and Higher Categories, Manchester, UK.

2023 **2-Sketches of Deduction**,
Category Theory at Work in Computational Mathematics and Theoretical Informatics, Bergen, Norway.

2022 **The geometry of coherent topoi and ultrastructures**,
Invited speaker, YaMCATS, Manchester, UK.

2022 **Bi-accessible and bipresentable 2-categories**,
Invited speaker, WG6 kick-off meeting: Syntax and Semantics of Type Theories, Stockholm, Sweden.

2021 **Formal Model Theory and Higher Topology**,
Seminar Lecture, Yorkshire and Midlands Category Theory Seminar (YaMCATS), online.

2020 **Formal Model Theory and Higher Topology**,
HoTT Seminar, Carnegie Mellon University, Pittsburgh, USA.

2019 **Topos theoretic approaches to formal model theory**,
Seminar Lecture, Institute de recerche et informatique fondamentale, Paris, France.

2018 **The shape of water: Homotopical algebra and set functors**,
Seminar Lecture, Max Plank institut, Bonn, Germany.

Selected Talks at recent conferences

2025 **Call doctrines by your name**,
Masaryk University, CT2025. Plenary speaker.

2024 **Topoi with enough points**,
University of Gothenburg, Logic Colloquium 2024.

2023 **The geometry of coherent topoi & ultrastructures**,
UCLouvain, CT2023.

2022 **Bi-accessible and bipresentable 2-categories**,
Masaryk University, Peripatetic Seminar on Sheaves and Logic 106.

2021 **Enriched Locally Generated Categories**,
University of Genova, Category Theory 2020->2021.

2021 **Formal Model Theory and Higher Topology**,
Online, Logic Colloquium 2021.

2021 **Towards higher Topology**,
Online, Toposes Online.

Administrative Duties

2025/now **Coordinator of the Master's program in logic at Gothenburg University**,
The Coordinator oversees curriculum design, faculty coordination, and academic quality. They support students with advising, admissions, and research while ensuring compliance with university policies. Additionally, they act as the program's main representative, managing communications and external collaborations. *Since March 2025 I coordinate the Master in Logic.*

2025/now **Member of the Program Council for the Master in Logic (Programrådet)**,
This Council oversees curriculum development, quality assurance, and academic policies to ensure the program meets educational standards. *I serve in the Program Council for the Master in Logic since March 2025*.

2025/now **Member of the Education Group (Utbildningsgrupp)**,
The Education Group oversees teaching quality, curriculum development, and pedagogical innovation across the department. It focuses on broader educational strategies, faculty support, and accreditation, distinct from program-specific councils. *I serve in the Education Group since March 2025*.

2024/2025 **Member of the Postgraduate Education Group (FUG) at Gothenburg University**,
The Postgraduate Education Group (FUG) is an advisory body for postgraduate (third cycle) education at the department. FUG consists of representatives of the active supervisory boards, two representatives of the doctoral student board, and the responsible education administrator. Representatives are appointed by the Associate Head of Department for Doctoral Studies. *I served between 2024 and 2025.*

Thesis Reader (Granskare) at Stockholm University,
Stockholm University evaluation process for Bachelor and Master thesis is based on a double review system. A part of the evaluation comes from an opponent and another one comes from a reader (granskare). The granskare must assess whether the thesis is solid enough to be defended. During my stay in Stockholm I was granskare few times. 2022: Thea Li, The internal logic of a topos. 2023: Caroline Roos, Agda formalization of Lindenbaum-Tarski algebras. *I served between 2022 and 2023.*

Thesis Reader at Masaryk University,
Masaryk University evaluation process for Bachelor and Master thesis is based on a double review system. One review comes from the advisor of the student and another one comes from an internal opponent. Both the reviews must describe, discuss and finally score the thesis from A to F. During my PhD studies I was three times opponent for bachelor theses. 2019: Yuriy Dupyn, Algebraic Structures. 2018: Jan Jurka, Locally presentable categories. 2018: Vladislav Demenchuk, Elements of nonstandard analysis. *I served between 2017 and 2020.*

Training and Development

PIL101 at Gothenburg University,
Teaching and Learning in Higher Education 1: Basic Course (PIL101). The mandatory modules offer an introduction to the pedagogical context in higher education and students' learning in higher education settings. They also address key aspects shaping higher education in Sweden, such as equality, ethics, and the laws, regulations, and local policy documents that govern university teachers' work. The elective modules have specific focuses. For instance, they may concentrate on a particular teaching method or be tailored to the needs of a specific faculty. Ongoing.

2026 **PIL102 at Gothenburg University**,
Teaching and Learning in Higher Education 2: Subject Field Pedagogy (PIL102). The focus is on the organisation and planning of courses and modules. Contextual factors, goals, content, learning activities, teaching, evaluation, assessment methods and university students' independent studies as well as the impact of the choice of course design are included in the course. These topics are highlighted from different perspectives and the relationships between these are debated. Particular attention is devoted to the course syllabus document.

PIL201 at Gothenburg University,
Supervision in Postgraduate Programmes: Subject Field Pedagogy (PIL201). The course is intended to provide knowledge about the research supervision regulations and insights into various aspects of the interaction between supervisor and doctoral student, and to provide communicative skills training and opportunities for problematization based on different supervision situations. Ongoing.